

REPORT

Of

Solutions Architecture: Assessing Client Needs and Designing a Solution

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: 2410030143

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Program: B.Tech CSE — AI & ML

IILM University, Greater Noida

2026-27

CANDIDATE’S DECLARATION

I, Yatharth Rathi, do hereby declare that the internship report titled “Solutions Architecture: Assessing Client Needs and Designing a Solution” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The report documents my learning and practical work completed through the Forage Solutions Architecture Job Simulation.

I affirm that this report represents my own academic work and that information taken from the internship certificate and other supporting material has been appropriately represented. I have not knowingly submitted this report to any other institute for a degree or diploma requirement. I shall be responsible for any copyright or academic-integrity issue arising from material used in this report.

Signature: _____

Student Name: Yatharth Rathi

Roll No.: 2410030143

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and encouraged me during the completion of this internship learning experience and the preparation of this report.

I am thankful to Forage for providing the Solutions Architecture Job Simulation, which offered a structured online environment to practise professional problem-solving. According to my Certificate of Completion, the simulation included the practical task “Assess the Client’s Needs and Design a Solution.”

I am also grateful to IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for providing an academic environment in which I could connect my B.Tech CSE — AI & ML studies with practical technology and business concepts.

Finally, I express my gratitude to my parents, teachers, and everyone who encouraged me throughout this learning experience.

Signature: _____

Student Name: Yatharth Rathi

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3. INTERNSHIP COMPLETION CERTIFICATE

The Certificate of Completion issued by Forage confirms Yatharth Rathi's completion of the Solutions Architecture Job Simulation on September 10th, 2026.



Inspiring and
empowering
future professionals

Yatharth Rathi Solutions Architecture Job Simulation

Certificate of Completion
September 10th, 2026

Over the period of September 2026, Yatharth Rathi has completed practical tasks in:
Assess the Client's Needs and Design a Solution

A stylized, handwritten signature in black ink.

Tom Brunskill
Co-Founder of Forage

Enrolment Verification Code 6aa2e4e29ad181a9cbafa243 | User Verification Code 6aa2e2a47b6283118cd770f0 | Issued by Forage

4. PROJECT DESCRIPTION

4.1 Introduction

This internship report presents a study of the Forage Solutions Architecture Job Simulation completed by Yatharth Rathi in September 2026. The experience was designed as an online, practical simulation in which the learner worked through a client-oriented technology problem.

Solutions architecture is concerned with translating business requirements into a practical technology solution. It requires an understanding of the client's objectives, constraints, users, existing environment and expected outcomes before recommending technical components.

The certificate supplied for this internship identifies the practical task as "Assess the Client's Needs and Design a Solution." This report therefore focuses on the process of understanding client needs, defining requirements, evaluating solution choices and communicating a high-level design.

The internship was a simulation-based learning experience rather than employment or a live production deployment. Its value lies in practising the reasoning and communication patterns expected in a professional solutions-architecture setting.

4.2 Organization Profile

Forage is an online virtual work-experience platform that provides structured job simulations. Its simulations are designed to let learners practise professional tasks in a guided environment and receive evidence of completion. The certificate used in this report was issued by Forage.

The Solutions Architecture Job Simulation provided the context for this report. The experience emphasised a client-first approach: understand the problem first, identify what the solution must accomplish, and then design a suitable architecture.

4.3 Problem Statement

A technology solution can fail to create business value when it is designed without a clear understanding of the client's actual needs. A client may have multiple objectives, technical limitations, budget constraints, security concerns and expectations for future growth. These factors must be considered together rather than independently.

The problem addressed by the simulation was therefore to practise a complete client-to-solution process: understand the client's situation, identify important requirements, evaluate appropriate solution approaches, and design and communicate a solution that is realistic and aligned with the stated needs.

A further challenge is communication. Even a technically sound architecture must be explained in a way that allows business and non-specialist stakeholders to understand its purpose, benefits, limitations and major trade-offs.

4.4 Project Objectives

- To understand the role of a solutions architect in a client-facing technology context.
- To practise assessing a client's business goals, requirements and constraints.
- To translate business needs into clear technical requirements.
- To consider scalability, security, reliability, performance and cost when evaluating a solution.
- To develop a high-level architecture that addresses the client's stated needs.
- To practise explaining technical recommendations in clear, structured language.
- To improve analytical thinking, problem-solving and professional communication.

4.5 Scope of the Project

The scope of the internship covered the client-facing stages of solutions architecture. It included understanding the client's needs, identifying requirements, considering constraints, designing a high-level solution and communicating the recommendation.

The project did not involve deployment of a production system, operation of live infrastructure, or ownership of a real client environment. The work should therefore be understood as a professional-learning simulation.

Within the simulation, the focus was on architecture reasoning rather than implementation-level coding. The learner's task was to think about what the solution should contain, why each major component or decision would be appropriate, and how the design would satisfy the client's needs.

The scope also included professional communication. A solution architecture is useful only when stakeholders can understand how it addresses the original problem and what trade-offs are involved.

4.6 Technologies and Tools Used

- Online simulation environment: Forage Solutions Architecture Job Simulation.
- Documentation: Written notes and structured report preparation.
- Architecture representation: High-level solution/architecture diagrams and structured component descriptions.
- Communication: Business-oriented explanation of technical recommendations.

The supplied certificate does not identify specific cloud providers, programming languages, modelling frameworks or software packages as required technologies. Accordingly, this report does not claim use of any particular implementation technology beyond the simulation environment itself.

4.7 Solution Architecture

The solution-architecture approach can be represented as a sequence of connected layers. The first layer is the Client and Business Need: understanding what the organisation is trying to achieve and what problem needs to be solved.

The second layer is Requirements and Constraints. Business objectives are translated into functional and non-functional requirements. Constraints such as security, budget, performance, reliability, scalability and implementation complexity are considered at this stage.

The third layer is Solution Design. Candidate technologies and architectural components are selected based on how well they satisfy the requirements. The design should avoid unnecessary complexity and should make important assumptions and trade-offs visible.

The fourth layer is Communication and Recommendation. The architecture is presented in a form that enables stakeholders to understand the proposed solution, its expected benefits, important risks and the reasoning behind the main design decisions.

A simplified architecture flow is shown below:

1. Client Needs	2. Requirements	3. Solution Design	4. Recommendation
Business goals Users Constraints	Functional needs Security Scale Cost	Components Integration Data flow Trade-offs	Benefits Risks Next steps Stakeholder decision

4.8 Methodology

The internship followed a client-first methodology. The process began by understanding the client's situation instead of immediately selecting a technology. This reduced the risk of designing a solution around a tool rather than around the actual problem.

Stage	Focus	Description
1	Understand the Client	Identify business objectives, users, pain points and constraints.
2	Assess Requirements	Separate functional requirements from non-functional requirements such as security, scalability and performance.
3	Explore Solution Options	Consider possible architecture patterns and technologies that could satisfy the requirements.
4	Evaluate Trade-offs	Compare alternatives using factors such as cost, complexity, reliability, security and future growth.
5	Design the Solution	Create a high-level architecture showing the major components and relationships.
6	Communicate	Explain the architecture, benefits, assumptions, risks and next steps in stakeholder-friendly language.

4.9 Expected Outcomes

- Improved understanding of the responsibilities and mindset of a solutions architect.
- Ability to begin technology design from client needs and business objectives.
- Better understanding of functional and non-functional requirements.
- Practical experience in considering architecture trade-offs instead of selecting technology in isolation.
- Improved ability to represent a solution at a high level and explain its components.
- Stronger communication skills for presenting technical recommendations to mixed audiences.
- Greater awareness of the importance of security, scalability, reliability, performance and cost in architecture decisions.
- A completed, verifiable Forage Certificate of Completion for the Solutions Architecture Job Simulation.

4.10 Certificate of Completion and Supporting Proof

The supplied Forage certificate confirms that Yatharth Rathi completed the Solutions Architecture Job Simulation on September 10th, 2026. It specifically records the practical task “Assess the Client’s Needs and Design a Solution.” The original certificate is attached in Chapter 3 of this report.

5. BIBLIOGRAPHY / REFERENCES

1. Forage, “Solutions Architecture Job Simulation,” Certificate of Completion issued to Yatharth Rathi, September 10, 2026.
2. Forage, Solutions Architecture Job Simulation learning materials and task instructions, as used during the online simulation.
3. IILM University, School of Computer Science and Engineering — academic programme context supplied by the student.

Note: This report intentionally limits specific technical claims to information supported by the supplied certificate and the simulation description available in the provided materials.